```
const int trigPin = 7;
const int echoPin = 8;

long duration;
int distance;

void setup()
{
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);

  Serial.begin(9600);
}

void loop()
{
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);

  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH);

  distance = duration * 0.034/2;

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.print("cm\n");

  delay(500);
}
```

```cpp
#include<Servo.h>

Servo myServo;
const int trigPin = 9;
const int echoPin = 10;

long duration;
int distance;

void setup() {
  myServo.attach(11);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);

  Serial.begin(9600);
}
void loop() {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);

  digitalWrite(trigPin, LOW);

  duration = pulseIn(echoPin, HIGH);
  distance = duration * 0.034/2;

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.print("cm\n");

  int angle = map(distance, 0, 50, 0, 180);
  angle = constrain(angle, 0, 180);

  myServo.write(angle);

  delay(100);
}
```

```
int irSensor = 9;
int led = 13;

void setup()
{
  Serial.begin(9600);
  Serial.print("Serial Working\n");

  pinMode(irSensor, INPUT);
  pinMode(led, OUTPUT);
}

void loop()
{
  int sensorStatus = digitalRead(irSensor);

  if(sensorStatus == 1){
    digitalWrite(led, LOW);
    Serial.print("Motion Ended.\n");
  }
  else{
    digitalWrite(led, HIGH);
    Serial.print("Motion Detected!\n");
  }

}
```